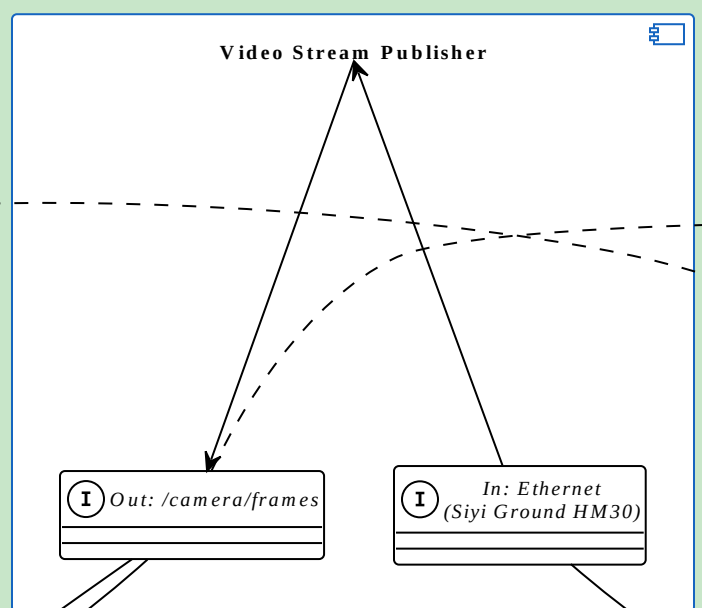
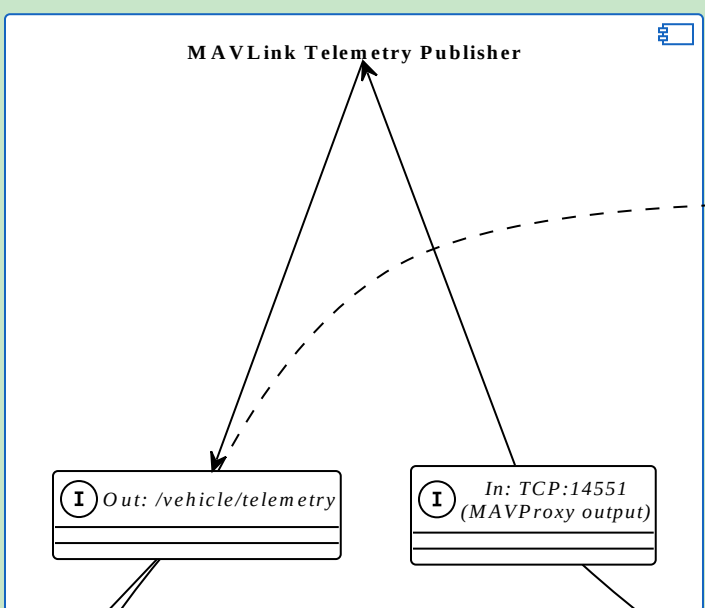
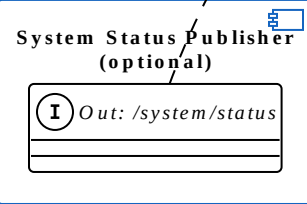


ROS2 Middleware Architecture & Message Flow
Custom GCS Software Communication

ROS2 Humble Middleware (Domain ID: 0)

Publishers (Data Sources)

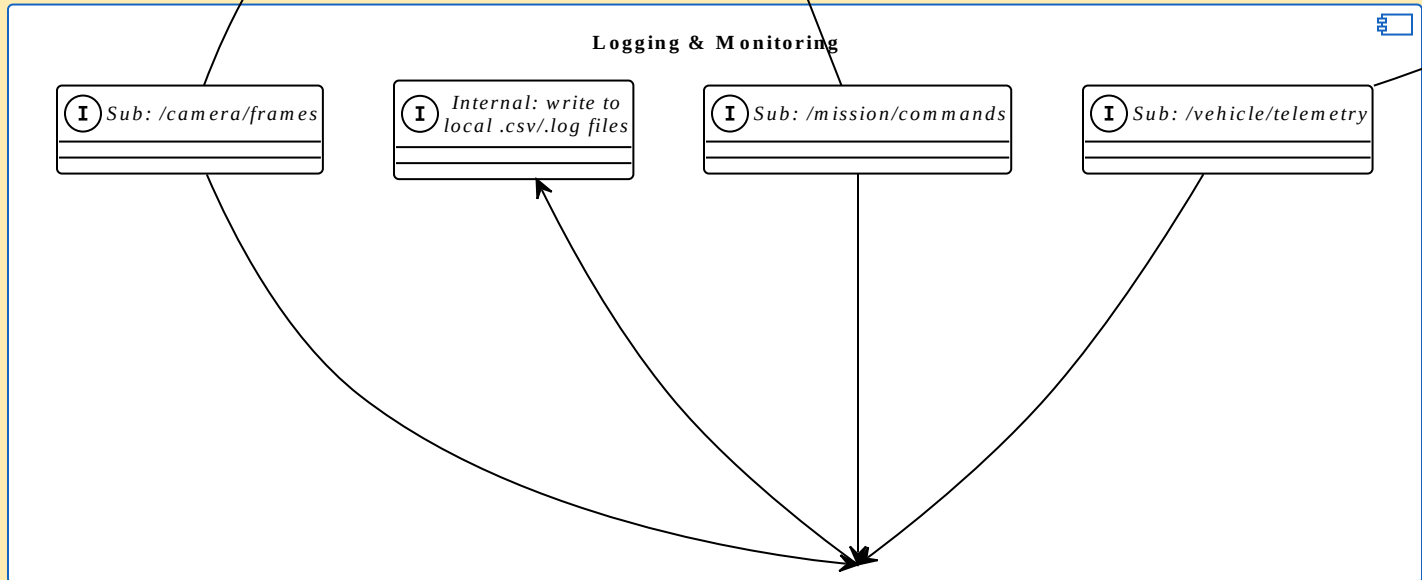
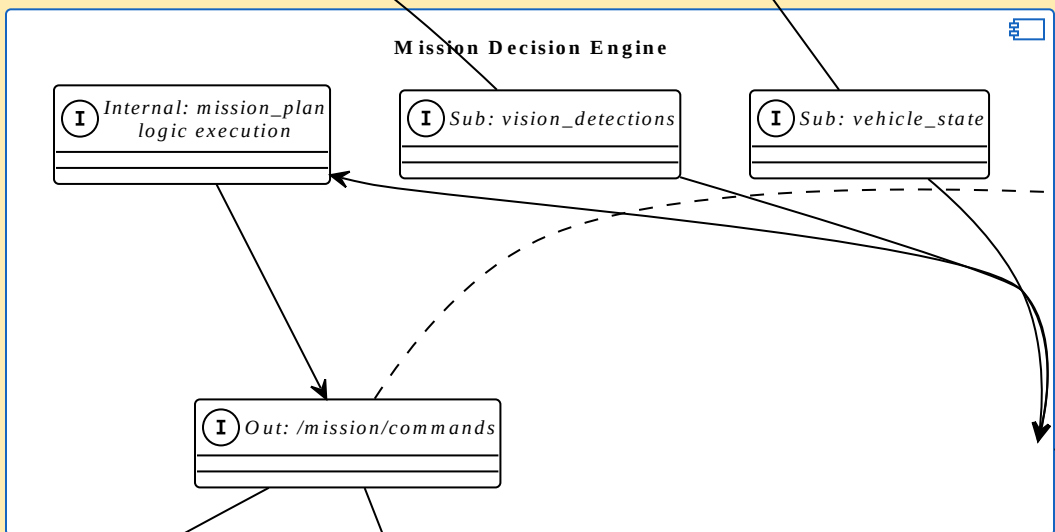
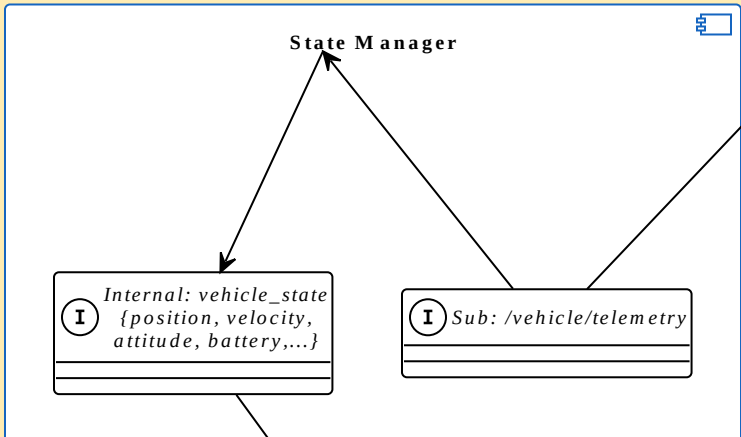
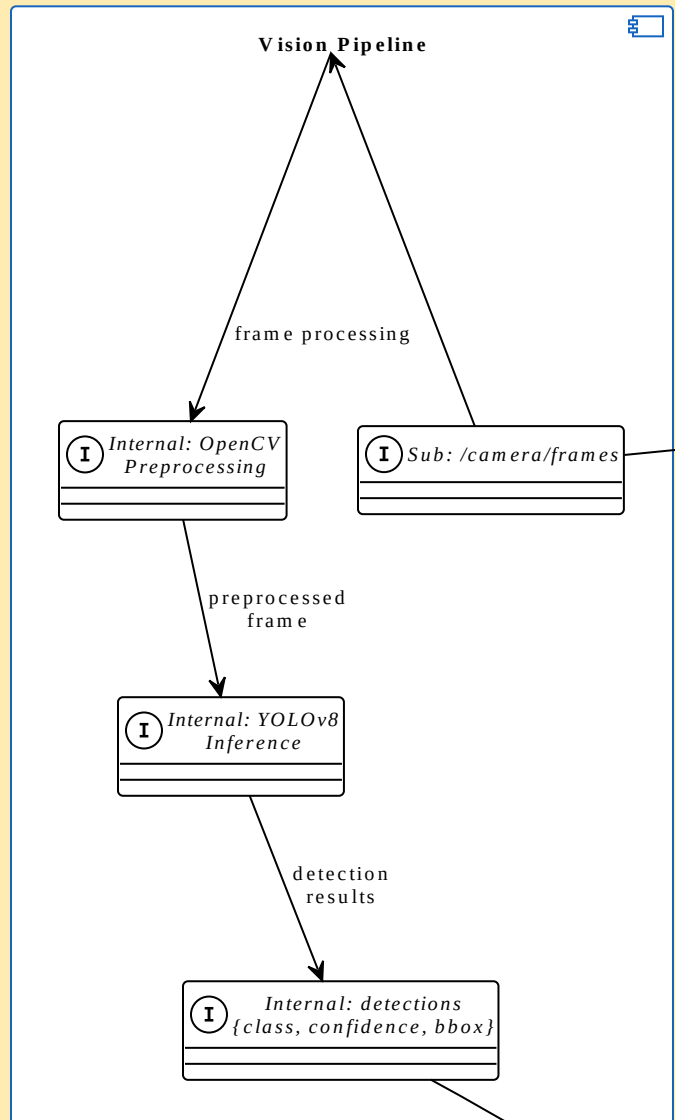


Topic: /system/status(Optional)
Message Type: Custom diagnostics_msgs
Fields:
• mission_state (enum)
• last_vision_detection (time)
• cpu_usage (float)
• memory_usage (float)
• inference_latency (float)
• command_latency (float)
Frequency: 1 Hz
QoS: BEST_EFFORT

Topic: /vehicle/telemetry
Message Type: geometry_msgs/TwistStamped + Custom
Fields:
• timestamp (float64)
• position.lat, lon, alt (float64)
• velocity.x, y, z (float32)
• attitude.roll, pitch, yaw (float32)
• attitude.rate.p, q, r (float32)
• battery.voltage (float32)
• battery.current (float32)
• battery.remaining (float32)
• gps.fix_type (uint8)
• armed (bool)
• flight_mode (string)
Frequency: 5-10 Hz (MAVLink 50Hz → ROS2 10Hz after filtering)
QoS: BEST_EFFORT

Topic: /camera/frames
Message Type: sensor_msgs/CompressedImage
Fields:
• header.stamp (time)
• header.frame_id (string)
• format (string) - "jpeg" or "h264"
• data (uint8[]) - compressed image
Frequency: 20-30 Hz
QoS: BEST_EFFORT (allow drops)
Size: ~50-150 KB/frame (compressed)

Custom GCS Software (Mission Logic Engine)



Subscription

Subscription

ROS2 Topic Subscription

ROS2 Topic Subscription

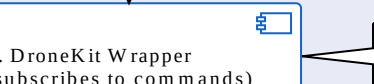
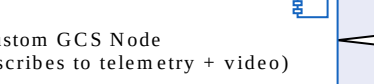
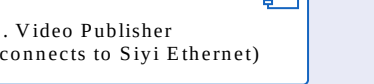
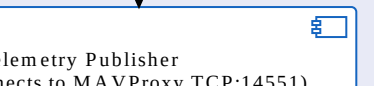
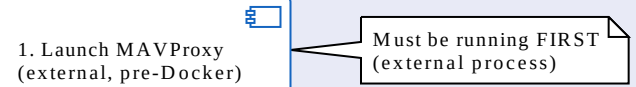
Subscribe telemetry

Receive video stream

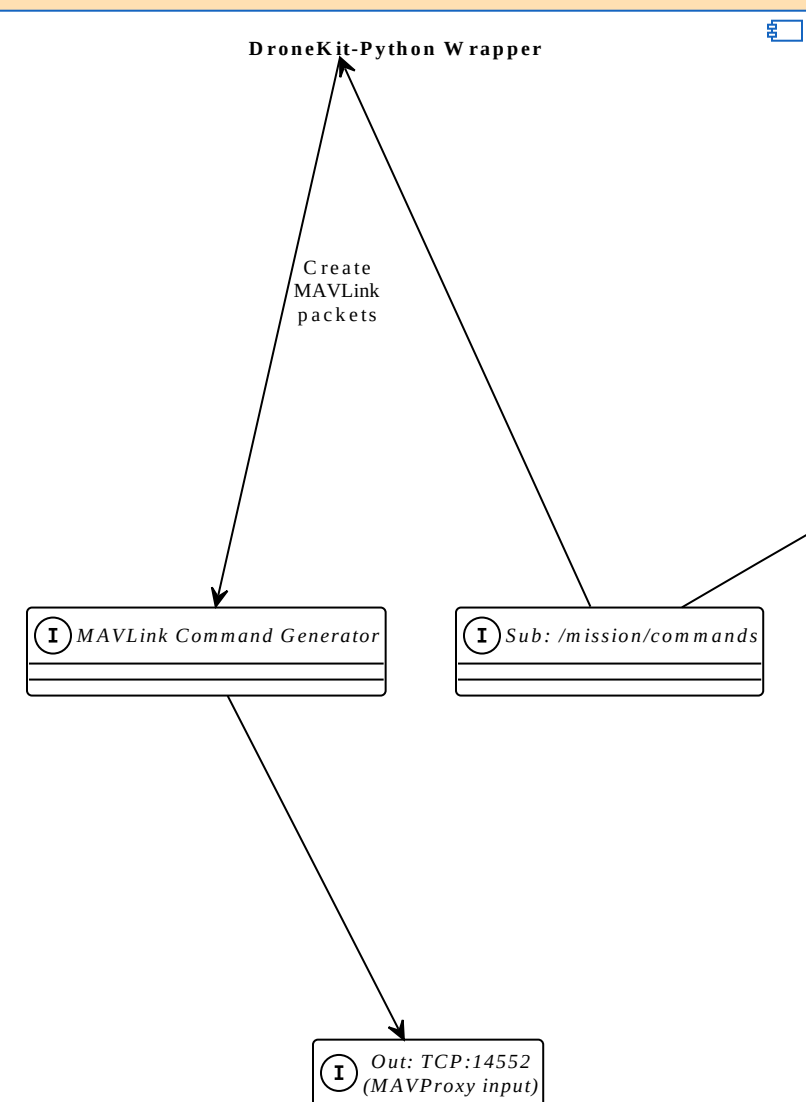
Topic: /mission/commands
Message Type: geometry_msgs/TwistStamped
Possible Command Types:
• set_mode (GUIDED, AUTO, LAND, ...)
• arm (true/false)
• set_position_target (lat, lon, alt, heading)
• set_velocity_target (vx, vy, vz)
• set_attitude_target (roll, pitch, yaw, throttle)
• trigger_payload (action_id, param)
Frequency: On-demand (0-10 Hz max)
QoS: RELIABLE

Mission Logic Pseudocode:
LOOP (every 100-500ms):
1. Read latest vehicle_state
2. Read latest vision_detections
3. IF vision_detections.empty:
→ Continue current mission
4. ELSE IF (detected_object matches_target):
→ Calculate intercept point
→ Update position_target
→ Publish SET_POSITION_TARGET command
5. ELSE:
→ Log detection
→ Continue search pattern
6. Check safety constraints:
→ battery_level > minimum
→ altitude < max_altitude
→ distance < max_range
7. IF constraint_violated:
→ Publish RTH command
Exit mission loop
END LOOP

ROS2 Node Startup Order

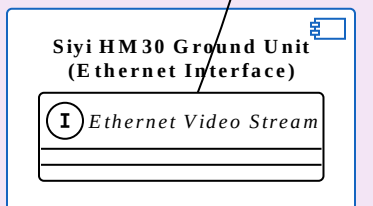
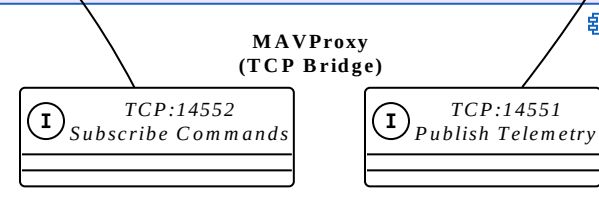
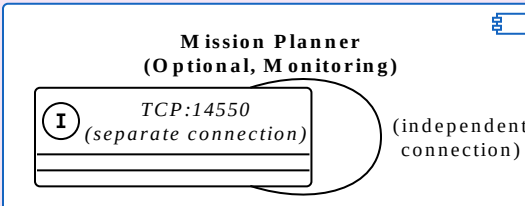


Command Output (Dronekit Interface)



Local Logging (Docker Container /data mount):
Files created:
• mission_YYYYMMDD_HHMMSS.csv
Columns: timestamp, lat, lon, alt, vx, vy, vz
• vision_YYYYMMDD_HHMMSS.log (timestamp, detections[], inference_time)
• commands_YYYYMMDD_HHMMSS.log (timestamp, command_type, params)
• gcs_YYYYMMDD_HHMMSS.log
General GCS events and errors

External Systems (Outside ROS2)



Processing Latency Breakdown

